CSE 2023 - HW2

1) a) $A \cap B$ b) $A - B$ c) $A \cup B$ d) $(A \cap B)'$

2) a) $B \subseteq A$ b) $A \subseteq B$ c) $A \cap B = \emptyset$ d) We can't say about that
e) $A = B$

3. a) $A = \{0, 1\}$, $B = \{1\}$, $C = \{0\}$, $A \neq B$

b) $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{2\}$, $A \neq B$

c) $A = \{1, 2, 3\}$, $B = \{1, 2, 3\}$, $C = \{1\}$, $A = B$

4.

A	B	C	$A \oplus B$	$B \oplus C$	$(A \oplus B) \oplus C$	$A \oplus (B \oplus C)$
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
1	0	0	1	0	1	1
0	1	1	1	0	0	0
1	1	0	0	1	0	0
1	0	1	1	1	0	0
1	1	1	0	0	1	1

Yes.

5.

A	B	C	$A \oplus C$	$B \oplus C$	
0	0	0	0	0	$A = B$
0	0	1	1	1	$A = B$
0	1	0	0	1	
1	0	0	1	0	
0	1	1	1	0	$A = B$
1	1	0	1	1	
1	0	1	0	1	
1	1	1	0	0	$A = B$

6. a) Yes b) No $f(1)=f(-1)=2$ c) Yes d) No $f(1)=f(-1)=2/3$

7. $f(x)=e^x$ $f^{-1}(x)=\ln x \iff x>0$ $x \in R \rightarrow f'(x) \neq \ln x$

8. $fog = (x+2)^2+1 = x^2+4x+5$

$gof = x^2+3$

9. a) countably infinite $11, 12, 13, 14, \ldots$

b) countably infinite $-1, -3, -5, \ldots$

c) finite

d) uncountable

e) countably infinite

f) countably infinite

10. a) $A=R$ $A-B=\{0\}$ finite
$B=R-\{0\}$

b) $A=(0,1)\cup N$ $A-B=N$ countably infinite
$B=(0,1)$

c) $A=R$ $A-B=R^+$ uncountable
$B=R^-\cup\{0\}$